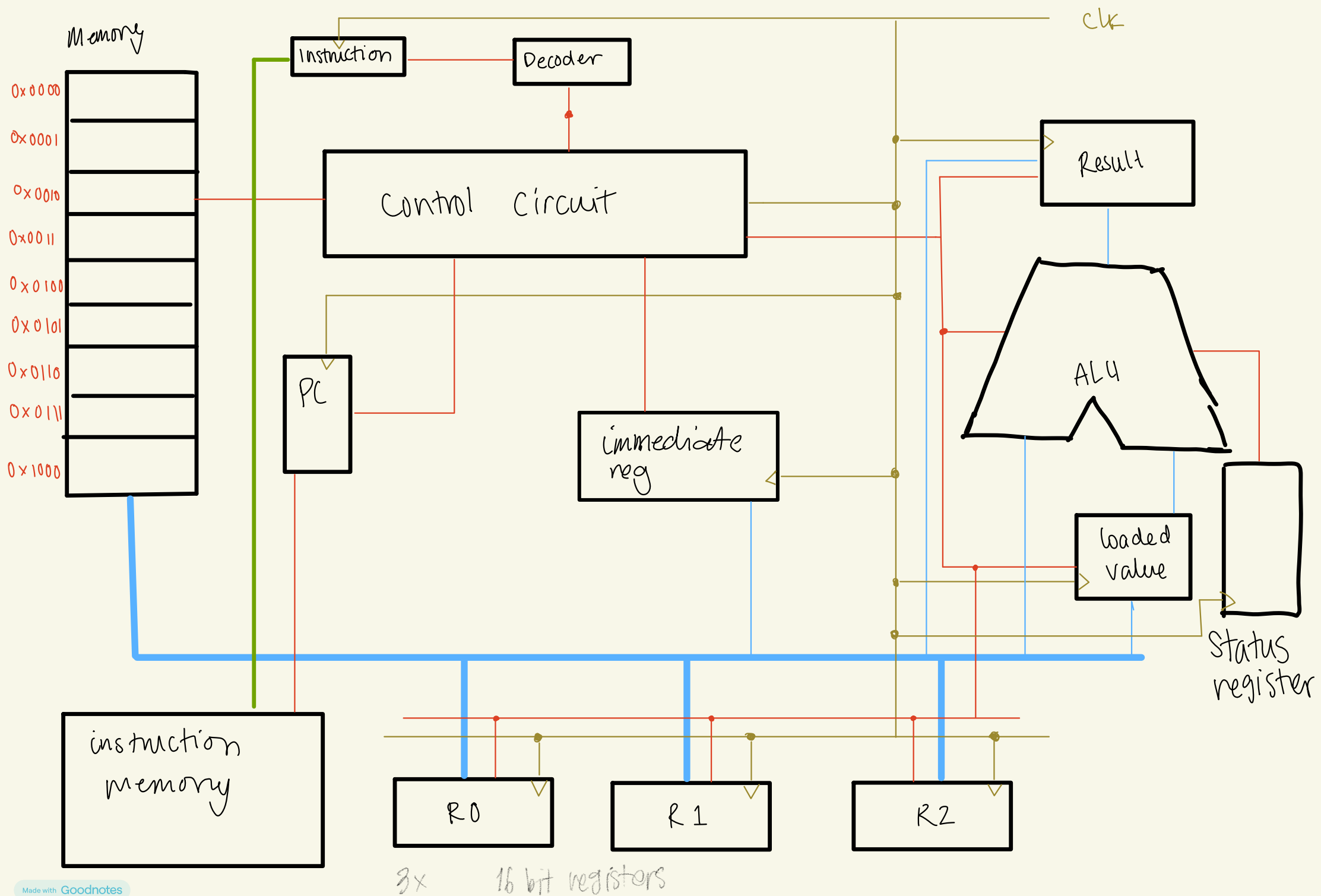
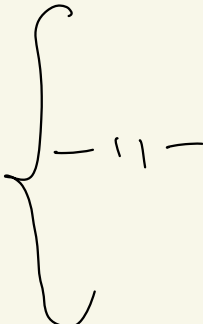
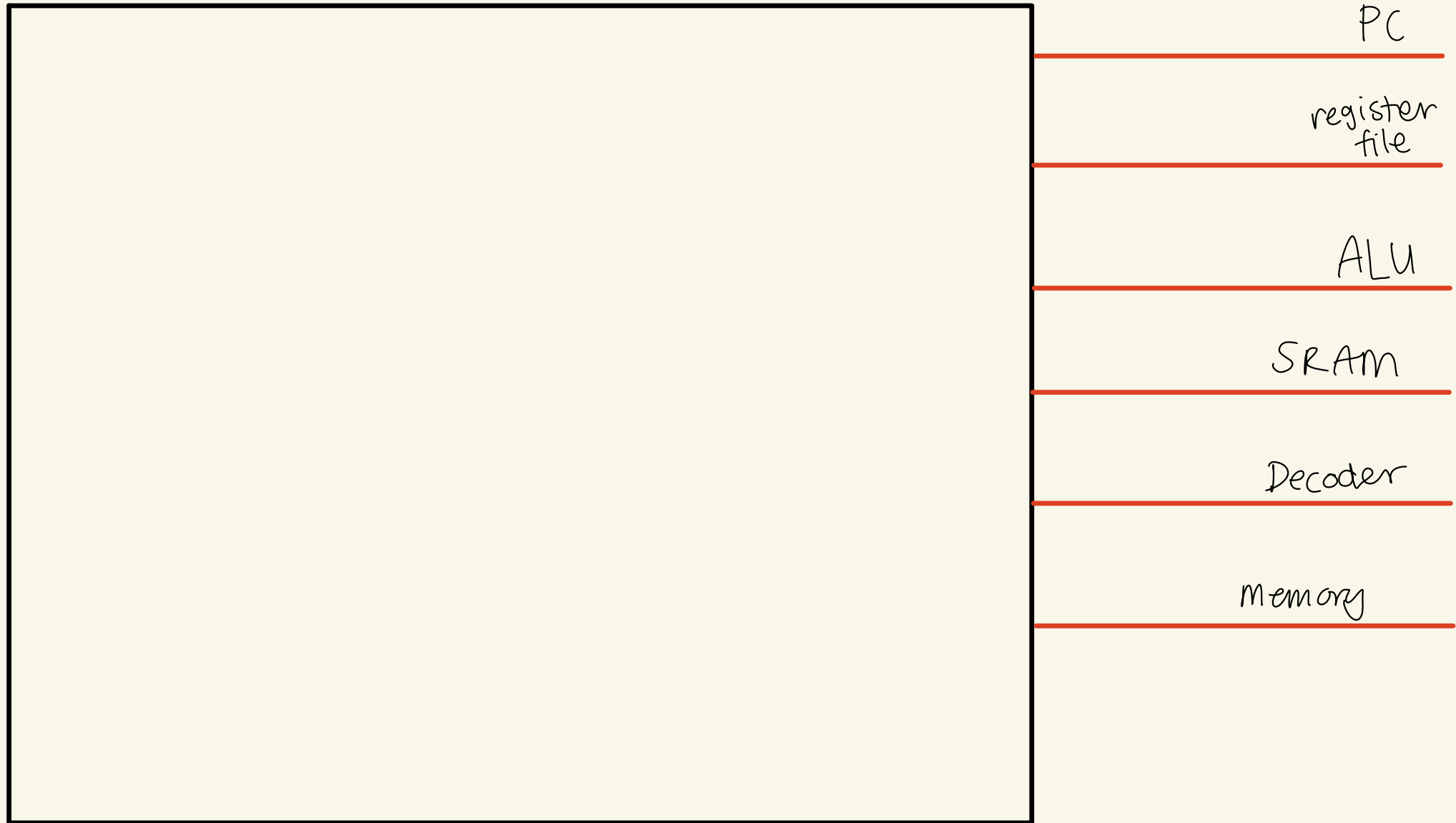




Opcode

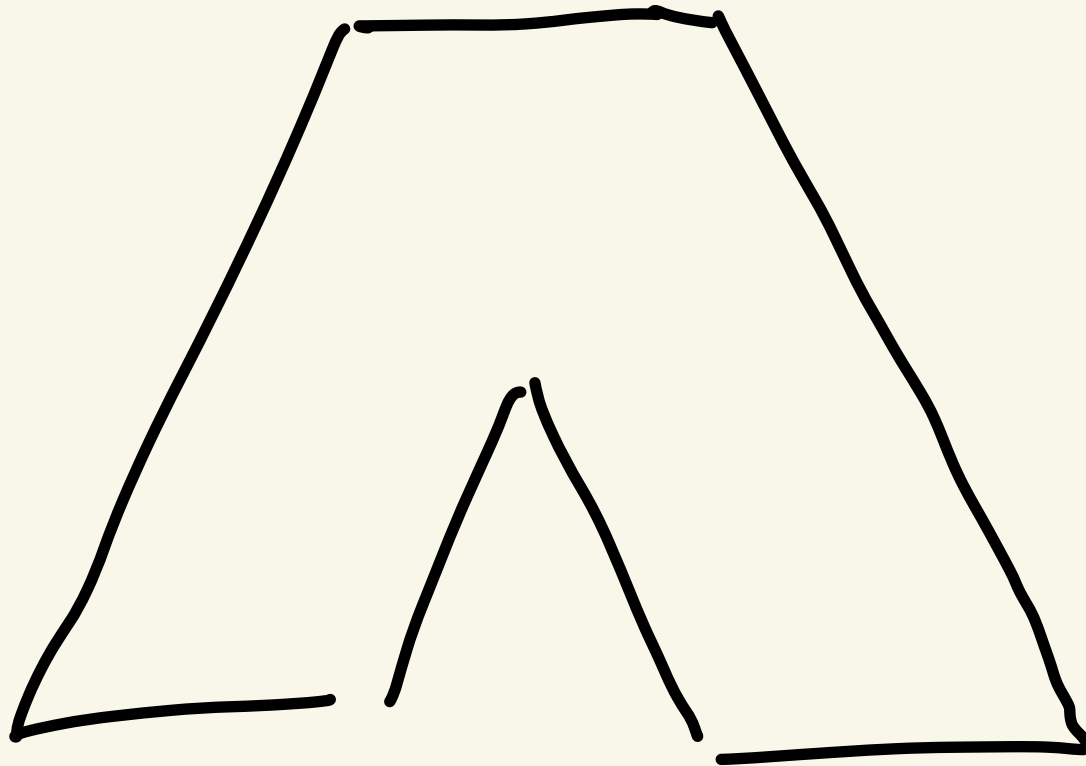
Instruction	opcode	register	data(H8)	data (L8)
LD	0x01	R[n:0]	memory address	
ST	0x02	R[n:0]	memory address	
ADD	0x11	R[n:0]	R[n:0]	R[n:0]
SUB	0x12	R[n:0]	R[n:0]	R[n:0]
LSL	0x13	R[n:0]	R[n:0]	R[n:0]
LSR	0x14	R[n:0]	R[n:0]	R[n:0]
AND	0x15	R[n:0]	R[n:0]	R[n:0]
OR	0x16	R[n:0]	R[n:0]	R[n:0]
XOR	0x17	R[n:0]	R[n:0]	R[n:0]
NOT	0x18	R[n:0]	R[n:0]	R[n:0]
CP	0x21		R[n:0]	
JMP	0x22		program counter addr	
BREQ	0x23		— 11 —	
BRNE	0x24			
BRLO	0x25			
BRSH	0x26			

Control circuit



ALU

Supported instructions



- Adder

↳ add

↳ 2's complement

1. flip bits

2. add 1

2 clock cycles

- Shift register (16 D flip-flops)

↳ LSL

↳ LSR

} ask Rudy!

- Bitwise operators

↳ OR, AND, XOR, NOT

↳ simply a truth table

- Comparison

↳ Status register

↑
again ask Rudy